Title of the Final Year Undergraduate Project

An undergraduate project report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

Degree of the Bachelor of the Science of Engineering Honours

by

R. Rathnaweera	-	RU/E/2005/004
A. Abekon	-	RU/E/2005/005
B. Bandara	-	RU/E/2005/006
C. Kankalu	-	RU/E/2005/007

....
Prof. A.B.C. Dee
(Supervisor)

....
Prof. A.B.C. Dee
(Co-Supervisor)

Abstract

Abstract goes here.

Acknowledgments

Acknowledgments goes here.

Contents

List of Figures

List of Tables

Acronyms

AC	-	Alternating Current
DC	-	Direct Current
LV	-	Low Voltage
MV	-	Medium Voltage

Chapter 1

Introduction

Small paragraph to make reader interested in further reading [1].

1.1 Background

Write background of your project.

1.2 Problem Statement

1.3 Objectives and Scope

1.3.1 Objectives

- Objective 1
- Objective 2
- ..

1.3.2 Scope

- Scope 1
- Scope 2
- ..

1.4 Literature Review

1.5 Methodology

1.6 Time Plan and Resources Required

Chapter 2

Report

All groups should submit a complete but fairly concisereport on the project. This very same document is also written in the format expected by the DEIE under UGPs [2].

2.1 Contents

Report should contains the following.

- Title Page
- Abstract
- Preface
- Acknowledgments
- Table of Content
- Acronyms (optional)
- List of Tables
- List of Figures
- Chapters
  - Introduction
  - Other Chapters
  - ...
  - Conclusions
- Appendices
- Bibliography

2.1.1 Introduction - The First Chapter

The Chapter 1, ”Introduction” may include similar to following contents.

- Background
- Problem Statement
- Objectives and Scope
- Methodology
- Time Plan and Required Resources

2.1.2 Other Chapters

Must be well organized to systematically provide essential information regarding the project [1, 3, 4].

2.1.3 Conclusions - The Last Chapter

Final chapter of the report usually contains the summary of results and logical conclusions together with the suggestions for future developments.

2.1.4 Appendices

An appendix is a section at the end of a book which includes supplementary information which may be of interest or use to the reader. Appendices are also used to support the qualifications of the author, and to increase the credibility of the publication [5].

2.1.5 Bibliography

Important points to remember on Bibliography.

- You may use alternative name References.
- List down all bibliography items.
- Each bibliography must be referred to in the text at appropriate places.
- refer bibliography by using number.
  ```
  At present 457 out of best 500 super computers use operating systems
  of Linux family [2].
  ```

2.2 Format

- Every formating aspect must have a very good reason.
- All formatting aspects must be perfectly consistent through out the report.

2.2.1 Documents

Divided into chapters, chapters are numbered using arabic letters (1, 2, 3,..), each chapter starts at a new page Chapters can be devied into sections (eg. for chapter 1 sections 1.1, 1.2, 1.3 ..). Sections can be divided into subsections (eg. 1.1.1, 1.1.2, 1.1.3, ..). Subsections are also can be divided into sub-subsections but do not use numbering [6].

Subsubsection

This way you may have subsubsections.

2.2.2 Page

A4, Portrait, all margins 1 inch, Gutter 0.25in, Header 0.25in, Footer 0.25in, Single Sided
Numbered consecutively starting from the first page of the main text through to the end of the report. Use small roman letters (i, ii, iii, iv) starting with the Preface (except Title Page and Abstract).

2.2.3 Font

Text: Times New Roman, Normal,12pt
Other effects such as underline, italic, .. can be used as appropriate.

2.2.4 Paragraph

Single Line Spacing, Justified, First line indented by 0.5in (except the first paragraph of the chapter)

2.2.5 Equations

All equations must be centered and aligned when multi-line. Number them right justified as (Chapter Number.Equation Number).
When necessary cite equations as for example Eq.(2.1) and Eq.(2.2).

$$\begin{aligned} H_c = \frac{1}{2n} \sum_{l=0}^{n} (-1)^l (n-l)^{p-2} \sum_{l_1+\cdots+l_p=l} \prod_{i=1}^{p} \binom{n_i}{l_i} \\ \cdot [(n-l)-(n_i-l_i)]^{n_i-l_i} \cdot \left[(n-l)^2 - \sum_{j=1}^{p} (n_i-l_i)^2\right]. \end{aligned} \tag{2.1}$$

$$y = mx + c_1 + mx^2 + c_2 \tag{2.2}$$

2.2.6 Figures

Figures (graphs, block diagrams, .. etc) are independent objects by nature i.e. figures should be self explanatory and understood by readers. Each figure must be numbered

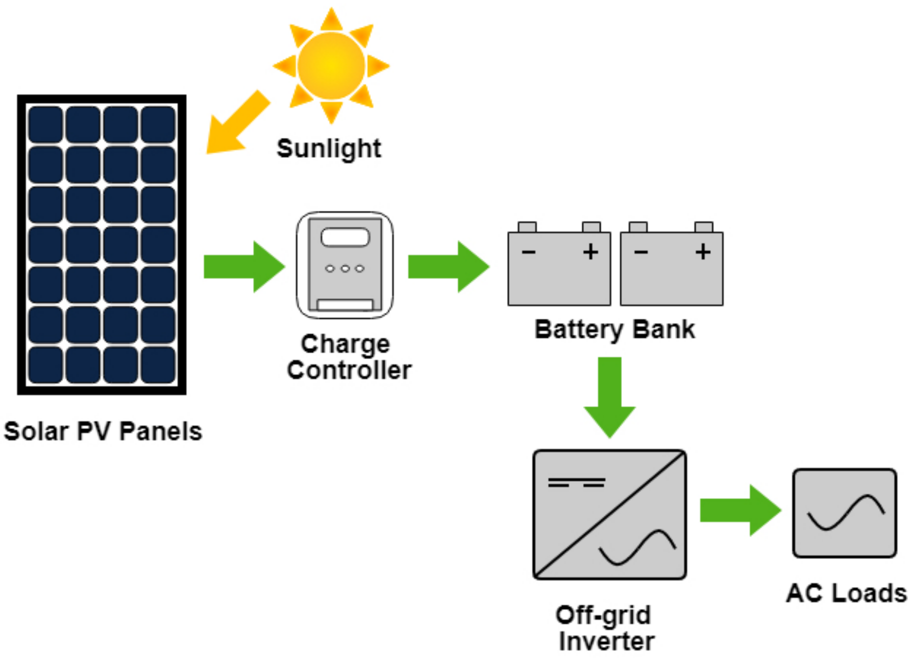

Figure 2.1: Off-grid PV system.

Table 2.1: This is an example for a table in latex.

Name	Age	Tel
Marasinghe X.Y.	22	0777-150-150
Singhemara X.Y.	33	0777-250-250

as Figure ChapterNumber.FigureNumber, cited and have a caption. Also should appear after citation. Good practice is to put figures in top or bottom of the page. Cite figures in the text as Figure 2.1.

2.2.7 Tables

Tables also are independent objects by nature. Each table must be numbered as Table ChapterNumber. TableNumber, cited and have a caption. Also should appear after citation. Good practice is to put tables in top or bottom of the page. Cite tables in the text as Table 2.1.

2.2.8 Bullets and Numbering

2.2.9 Numbering

1. Level 1
2. Level 1
   (a) Level2

   (b) Level 2
      i. Level 3
      ii. Level 3

3. Level 1

2.2.10 Bullets

- Level 1
- Level 1
  - Level2
  - Level 2
    - Level 3
    - Level 3
- Level 1

2.2.11 Description

Step 1 Open a text editor such as Emacs.

Step 2 Close WYSIWYG editors

Step 3 Goto Step 1.

2.2.12 Mixed

1. Level 1
2. Level 1
   - Level2
   - Level 2
     (a) Level 3
     (b) Level 3
3. Level 1

2.3 Notes on Writing

- Simplest possible English.
- Simple Present Tense (Passive Voice): Explain the project
- Present Perfect Tense (Passive Voice): Explain experiments
- Past Perfect Tense: Use only if you can not live without.
- Consistency (e.g. symbols used, format, ..)

Chapter 3

Assessment

3.1 Project Proposal

3.2 Project Progress

3.3 End Exam

Chapter 4

Conclusion

Appendix A

Essential English Grammar - A Handout

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

A.1 Simple Present Tense

A.2 Perfect Tense

A.3 Past Past Perfect Tense

A.4 Passive Voice

A.5 Technical Writing Fundamentals

A.6 Proof Reading

Appendix B

Graphical Interpretation of Data

The graphical interpretation given below shows the relationship between x and y is acceptable to the purpose of the project

B.1 Data Analysis

B.1.1 Statistical Analysis

B.1.2 Empirical Analysis

B.2 The X vs Y vs Z

Bibliography

[1] M. Goossens, F. Mittelbach, and A. Samarin, *The LaTeX Companion*. Reading, Massachusetts: Addison-Wesley, 1993.

[2] A. Abdelhakim, P. Mattavelli, and G. Spiazzi, “Split-source inverter,” in *IECON 2015 - 41st Annual Conference of the IEEE Industrial Electronics Society*, pp. 001288–001293, Nov 2015.

[3] G. D. Greenwade, “The Comprehensive Tex Archive Network (CTAN),” *TUGBoat*, vol. 14, no. 3, pp. 342–351, 1993.

[4] P. Hans, *EX2000 User Manual*. Chroma Inc, 1990.

[5] E. Hover, “Latex help manual,” tech. rep., EPRI, 2000.

[6] “How to Prolong Lithium-based Batteries.” http://batteryuniversity.com/learn/article/how_to_prolong_lithium_based_batteries, 2003. [Online; accessed 05-05-2017].